

Yuting Irene Li

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**Postdoctoral researcher, Hannezo Group,
Institute of Science and Technology Austria**

Career

- **2023 - present**
Postdoctoral researcher at Hannezo Group, ISTA
Independent funding from the IST-Bridge international postdoctoral program, part of Marie Skłodowska-Curie cofund Action.
- **2021 - 2023**
Research Associate at Simons Lab, University of Cambridge
Independent funding from Engineering and Physical Sciences Research Council, UK.
- **2017 - 2021**
PhD student at Trinity College, University of Cambridge with Jardine and Cambridge Trust scholarship
Supervised by Prof. M.E. Cates in the Soft Matter Group at the Department of Applied Mathematics and Theoretical Physics.
- **2016 - 2017**
Master of Maths at Trinity College, University of Cambridge
Specialising in applied maths and theoretical physics: quantum and statistical field theories, symmetry groups and condensed matter.
Graduated with Distinction (highest classification).
- **2013 - 2016**
BSci in Physics at Trinity College, University of Cambridge
First Class (highest classification) in all three years.

Awards and fellowships

- 2023 - 2025: **IST-Bridge** fellowship, part of the Marie Skłodowska-Curie cofund Action.
- 2020: **Royal Society** RAMP Early Career Investigator Award (RECIA) for exceptional contribution towards pandemic modelling.
- 2017 - 2021: **Jardine and Cambridge Trust** scholarship for PhD students
- 2017 - 2021: **Research Scholar** at Trinity College, Cambridge
- 2013 - 2017: **Junior Scholar** and then **Senior Scholar** at Trinity College, Cambridge.
- 2013: Silver medal at **International Physics Olympiad**. Representing the United Kingdom.

Talks and seminars

- 2025 invited talk and panelist at **Edwards Symposium** at Cambridge University.
- 2025 invited talk at the **Hubrecht Institute**.
- 2025 invited talk at **Patterns Across Scales** Symposium at Vienna.
- 2025 contributed talk at **French-German WE-Heraeus-Seminar** at Les Houches “From Soft Matter to Biophysics”
- 2024 contributed talk at **Company of Biologist** workshop on Physics of the Early Embryonic Cell Divisions
- 2024, 2023 **ISTA** Soft Hour seminars and FriSBI (Friday Systems Biology), various topics
- 2018, 2022 **University of Cambridge** Statistical Physics and Soft Matter Seminars and Lennard-Jones Centre seminars, various topics.
- 2020 **RAMP** presenting on behalf of the Cambridge Soft Matter group at the meeting between RAMP volunteers and SPI-M members on Covid modelling
- 2020, 2018 Invited seminars at **Imperial College** and **Warwick**
- poster presentations at various conferences including **Statphys27**, **Physics of Living Matter**, **Edwards Centre Symposium**, **EMBO Physics of Living Systems**, etc.

Teaching and outreach

- **Teaching assistant for graduate courses (2025)** Data science core track course at ISTA.
- **PhD rotation project supervision (2025)** Acting as day-to-day contact for a rotation student on a project on active gel theory.
- **Master project supervision (2022)** Supervising a physics master student in Cambridge for a research project on the Kuramoto Model.
- **Undergraduate supervisions** Supervising Maths third year courses including Principles of Quantum Mechanics (2017, 2018, 2019), Statistical Physics (2018, 2019, 2021) and Mathematical Biology (2020, 2021).
- **Pint of Science public outreach (2022)** Speaking at the Pint of Science event in Cambridge on Active Matter.
- **Outreach for A-level students (2019)** Teaching interview preparation classes at Women in STEM residential for Year 12 students hosted by Trinity College.
- **Outreach for Maths master students (2019)** Volunteering at events organised for female Maths Part III students.
- **Trinity Maths Society Symposium (2019)** Presenting at the Trinity Maths Society Centenary Symposium (an undergraduate maths society) on Active matter.

Publications (lead theory author on all papers except the last)

- Tong, X.*, Li, Y. I.*, Schelle, J., Hannezo, E., Heisenberg, C-P. (2025) Non-canonical cytokinesis driven by mechanical uncoupling via nematic flows and adhesion-based invagination. [revision at Nature Cell Biology, *bioRxiv*]
- Mishra, N., Li, Y. I., Hannezo, E., Heisenberg, C-P., (2026) Geometry-driven asymmetric cell divisions pattern cell cycles and zygotic genome activation in the zebrafish embryo, *Nature Physics*
- van Oostrom, M. V., Li, Y. I., ... Simons, B. D., Sonnen, I. (2025) Scaling of mouse somitogenesis by coupling of cell cycle to segmentation clock oscillations, [revision at Cell Stem Cell, *bioRxiv*]
- Martinet, Q.*, Li, Y. I.*, Aubret, A., Hannezo, E., Palacci, J. (2025) Emergent dynamics of active elastic microbeams, *Phys. Rev. X*
- Sato, T., Li, Y. I., ... Simons, B. D., Shosei, Y. (2024) Self-organization of spermatogenic wave coordinates sustained sperm production in the mouse testis, [revision at *Cell*]
- Li, Y. I., Garcia-Millan, R., Cates, M. E., Fodor, É. (2023) Towards a liquid-state theory for active matter. *Europhysics Letters*
- Li, Y. I., Cates, M. E. (2021). Hierarchical microphase separation in non-conserved active mixtures. *EPJ E*
- Li, Y. I., ..., Jack, R. L. (2021). Efficient Bayesian inference of fully stochastic epidemiological models with applications to COVID-19. *Royal Society Open Science*
- Li, Y. I., Cates, M. E. (2020). Steady state entropy production rate for scalar Langevin field theories. *J. Stat. Mech.: Theory Exp.*
- Li, Y. I., Cates, M. E. (2020). Non-equilibrium phase separation with reactions: a canonical model and its behaviour. *J. Stat. Mech.: Theory Exp.*
- Engel, E. A., Li, Y. I., Needs, R. J. (2018). First-principles momentum distributions and vibrationally corrected permittivities of hexagonal and cubic ice. *Phys. Rev. B*

Past Experiences

- Organiser for **FriSbi** (Friday Systems Biology) seminars at ISTA (2023 - 2025) and Soft Matter and Statistical Physics seminars at Cambridge (2020 - 2021).
- **2020: Pyross Project** Developing open source software for epidemiological modelling with Cambridge Soft Matter group as part of the RAMP initiative. <https://github.com/rajeshrinet/pyross>.

- **2017, 2016: Summer Student at Theory of Condensed Matter group in Cavendish Laboratory, University of Cambridge.** Working on using Matrix Product State for machine learning (2017), and anharmonic vibrational corrections to ice (2016).